

Q.1]

→ Equivalence class partitioning [ECP]

① alphabet / alphanumeric characters

• valid: only letters and digits → AB123

• invalid: contains special character → AB@123

② Length (3-15 characters)

• valid: 3-15 → ABC123

• invalid: <3 → AB

• invalid: >15 → AB12345678901234

③ First two character must be letter → AB123

• valid: First two are letters → AB123

• invalid: First two not letters → A1123, 1B123

• Boundary Value analysis (BVA) - Length:

Input	Length	Expected
AB	2 2	invalid
ABC	3	valid
ABCD	4	valid
AB12345678901	14	valid
AB1234567890123	15	valid
AB12345678901234	16	invalid

Q.2]

→ Equivalence class partitioning is weak at handling combinations of input conditions - it tests inputs independently and may miss error caused by interactions between conditions.

- cause effect modeling logical relationships (AND, OR, NOT) between input conditions.
- ensuring combinations of causes are tested, not just single conditions.

Q.3]

→ causes (inputs)

- C1: string length is within valid range.
- C2: string length is out of range.
- C3: character exists in the string.
- C4: character does not exist in the string.

effects (output)

- E1: error message displayed.
- E2: position of character displayed.
- E3: "not found" message displayed.

cause-effect relationships

- IF C2 → E1
- IF C1 AND C3 → E2
- IF C1 AND C4 → E3

Test cases (Decision Table)

Test case	C1	C2	C3	C4	expected o/p
TC1	No	Yes	-	-	error msg
TC2	Yes	No	Yes	No	position displayed
TC3	Yes	No	No	Yes	"not found" msg

Q.4]

→ cause effect graphing state transition testing

① Used to test combinations of inputs

① Used to test system behavior over time

② best for complex business rules

② best for event-driven systems.

③ does not consider previous state

③ depends on current & previous state

④ e.g. for Validation

④ e.g. login system

Q.5]

→ equivalence classes

• Valid equivalence classes

EC1: color = red

EC2: color = blue

EC3: color = yellow

EC4: color = violet

• Invalid

• EC5: color not in the given set.

Test case	Input color	accepted result.
TC1	red	accepted
TC2	green	Rejected
TC3	blue	Accepted
TC4	orange	Rejected.

Q. 6]

Assumption:

- amount is entered in dollars (no cents)
- amount must be positive ≤ 200 and a multiple of \$5.

• Equivalence class partitioning (ECP)

equivalence classes:

- EC1 (Valid): amount ≤ 200 and multiple of \$5
- EC2 (Invalid): amount ≤ 200 but not a multiple of \$5
- EC3 (Invalid): amount ≥ 200
- EC4 (Invalid): amount ≤ 0

Test cases

EC	Test input	Expected Result
EC1	100	Accepted
EC2	102	Rejected
EC3	405	Rejected
EC4	0	Rejected

• Boundary value analysis (BVA)

boundaries

- lower boundary: \$0
- upper boundary: \$200

Test cases

Test input	boundary case	Expected Result
-5	below min	Rejected
5	min valid multiple	Accepted
195	just below max	Accepted
200	max	Accepted
205	just above max.	Rejected

Q.7]

→ assumptions

- exactly 25 plotting-point numbers are provided.
- Valid bounds: $1.0 < \text{lower} < \text{upper} \leq 5000.0$
- numbers equals to the bounds are included.
- on Invalid bounds → error message + reprompt.

Equivalence class partitioning (ECP)

Equivalence classes.

- EC1 (Valid): $1.0 \leq \text{lower} < \text{upper} \leq 5000.0$
- EC2 (Invalid): $\text{lower} < 1.0$
- EC3 (Invalid): $\text{upper} > 5000.0$
- EC4 (Invalid): $\text{lower} > \text{upper}$

EC#	lower	upper	expected result.
EC1	10.0	100.0	sum, avg, count display
EC2	0.5	100.0	error message
EC3	100	6000.0	+
EC4	100.0	50.0	+

Numbers to check

equivalence classes

- EC5 (Valid): numbers within or on bounds
- EC6 (Invalid): numbers outside bounds

EC	numbers (e.g.)	expected result
EC5	Values between bounds	counted in sum & avg
EC6	Values outside bounds	Ignored.

Boundary Value Analysis (BVA)

boundary value analysis

test cases	lower	upper	expected Result
BVA1	1.0	5000.0	Valid computer result
BVA2	0.9	100.0	error
BVA3	1.0	1.0	error
BVA4	4999.9	5000.0	Valid
BVA5	1.0	5000.1	error

Q.8]

equivalence classes

boundary inputs

- EC1 (Valid): $1.0 \leq \text{lower} < \text{upper} \leq 5000.0$
- EC2 (Invalid): $\text{lower} < 1.0$
- EC3 (Invalid): $\text{upper} > 5000.0$
- EC4 (Invalid): $\text{lower} \geq \text{upper}$

numbers to any

- EC5 (Valid): numbers within or equal to bounds
- EC6 (Invalid): numbers outside bounds

boundary values

- lower boundary: 1.0
- Upper boundary: 5000.0
- just below / equal / just above bounds.

Test cases

TC ID	ECs covered	low	upper	Sample value	expected resp
TC1	EC1, EC5	10.0	100.0	Valid with base	sum and avg cont
TC2	EC2	6.5	100.0	any	error
TC3	EC3	10.0	5000.1	any	error
TC4	EC4	100.0	100.0	any	error
TC5	EC1, EC6	1.0	5000.0	5000.0	Valid with base
TC6	DATA	1.0	5000.0	5000.0	correct sum, avg cont

• Coverage summary

Class / boundary	Covered
EC1	✓
EC2	✓
EC3	✓
EC4	✓
EC5	✓
EC6	✓
Lower ≥ 1.0	✓
Upper = 5000.0	✓

Q.9]

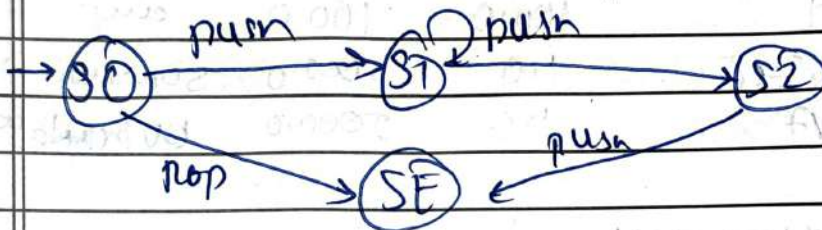
State of the stack machine.
Let stack capacity = n

S0: empty $\rightarrow$ stack has 0 items

S1: partially filled $\rightarrow$ stack has 1 to $n-1$ items.

S2: stack has n items.

S3: error $\rightarrow$ invalid operation.



Q.10]

Random Testing

- test inputs are generated randomly.
- No tester experience required.
- may miss critical defects.

Error Guessing

- test cases are based on tester's experience & intuition.
- focus on likely error areas.
- more effective at finding severe defects.